

# Assignment No. 9

Group 16

## Problem Statement

Identify potential risks in the **Road Traffic Analysis and Prediction System** and prepare appropriate mitigation strategies to ensure successful project execution.

## Abstract

Risk management is a critical aspect of software development that helps identify potential issues that may affect project success. Proper risk identification and mitigation strategies help reduce uncertainty, improve system reliability, and ensure smooth project execution.

In this assignment, various technical, operational, and security risks associated with the Road Traffic Analysis and Prediction System are identified. Suitable mitigation strategies are proposed to minimize the impact of these risks.

## Introduction

The development of a Road Traffic Analysis and Prediction System involves multiple complexities, including real-time data integration, complex prediction algorithms, and handling large volumes of crowdsourced data. Such systems are highly sensitive to data accuracy and service availability.

Risk management involves:

- Identifying risks
- Analyzing their impact
- Planning mitigation strategies
- Monitoring risks during development

This assignment focuses on identifying possible risks and providing effective solutions for the traffic project.

## Types of Risks

Risks in software systems can be broadly categorized into:

- Technical Risks (Algorithm complexity, API dependency)

- Security Risks (User data protection)
- Operational Risks (Data accuracy, server availability)
- Project Management Risks (Budget, requirement changes)
- External Risks (Third-party API changes)

## Risk Identification and Analysis

| ID  | Risk Description                | Type        | Prob.  | Impact |
|-----|---------------------------------|-------------|--------|--------|
| R1  | API Integration Failure (Maps)  | Technical   | Medium | High   |
| R2  | Data Inaccuracy (Crowd-sourced) | Operational | High   | Medium |
| R3  | Server Downtime                 | Technical   | Low    | High   |
| R4  | Unauthorized Data Access        | Security    | Low    | High   |
| R5  | Performance Bottlenecks         | Technical   | Medium | Medium |
| R6  | High Cost of API Usage          | Project     | Medium | Medium |
| R7  | Model Misclassification         | Technical   | Medium | Medium |
| R8  | Low User Adoption               | Operational | High   | Medium |
| R9  | Requirement Changes             | Project     | Medium | Medium |
| R10 | Deployment Issues               | Operational | Low    | Medium |

## Risk Mitigation Strategies

### R1: API Integration Failure

#### Mitigation:

- Implement fallback mechanisms (use historical averages if API fails).
- Implement circuit breakers for external service calls.
- Monitor API usage and health status regularly.

### R2: Data Inaccuracy

#### Mitigation:

- Implement verification algorithms (e.g., cross-referencing multiple reports).
- Use confidence thresholds for AI-based pothole detection.
- Allow users to "upvote" or "verify" existing reports.

## R4: Unauthorized Access

### Mitigation:

- Use JWT (JSON Web Tokens) for secure authentication.
- Implement Role-Based Access Control (RBAC).
- Encrypt sensitive user data in the database.

## R5: Performance Issues

### Mitigation:

- Optimize database queries and use indexing.
- Implement server-side caching for frequent prediction requests.
- Use efficient route sampling techniques.

## R6: API Cost Management

### Mitigation:

- Cache API results to minimize redundant calls.
- Implement usage limits per user to prevent abuse.
- Explore alternative open-source data sources (OSM).

## R7: Model Misclassification

### Mitigation:

- Regularly retrain the pothole detection model with new data.
- Implement a manual review queue for low-confidence detections.

## Risk Priority Matrix

| Impact        | Low Probability  | High Probability                  |
|---------------|------------------|-----------------------------------|
| High Impact   | Monitor (R3, R4) | Critical (R1)                     |
| Medium Impact | Accept (R10)     | Moderate (R2, R5, R6, R7, R8, R9) |

## Conclusion

Risk management is essential for the successful development of a reliable Road Traffic Analysis and Prediction System. In this assignment, various risks associated with the system were identified and analyzed. Appropriate mitigation strategies were proposed to minimize the impact of these risks, ensuring better system stability, security, and performance.